C O N S T E L L A T I O N

Concept Paper

The Constellation Map

A Radial Knowledge Visualization Engine

Inspired by Goalscape • Reimagined for Personal Knowledge Management

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Date: April 2026
Version 1.0 — Internal Working Document

1. Executive Summary

This concept paper proposes the **Constellation Map** — a radial knowledge visualization engine for Constellation, the PKM application. The concept draws direct inspiration from Goalscape, a goal management platform whose distinctive circular chart encodes hierarchy, weighted priority, and progress within a single visual frame.

Where Goalscape applies its radial chart to goal decomposition and project tracking, the Constellation Map reimagines the same structural principles for knowledge management: visualizing note density, knowledge maturity, conceptual coverage, and structural gaps across a user's universe/library. The result is a unique view that no major PKM tool currently offers — a single-glance **knowledge cartography** that reveals both the shape of what a user knows and the boundaries of what they have yet to explore.

Core proposition

The Constellation Map transforms Constellation's underlying note graph into an interactive radial chart where angular width represents knowledge weight, fill depth represents maturity, and concentric rings represent hierarchical depth — giving users an instant, holistic portrait of their knowledge landscape.

2. Source Analysis: Goalscape

2.1 What Goalscape is

Goalscape is a visual goal management platform developed by Marcus Baur, an Olympic sailor who needed a way to see all dimensions of athletic preparation in a single view. The product has been adopted by organizations including Volkswagen, Georgetown University, UK Sport, and the Province of Trentino (for Winter Olympics planning). It is based in Kiel, Germany, and is supported by EU and state-level AI innovation grants.

At its core, Goalscape replaces linear task lists and spreadsheets with a circular, multi-level chart — what the company calls a "goal map." This chart is not a standard sunburst visualization. It introduces several mechanics that distinguish it from generic hierarchical charts.

2.2 Core mechanics of the Goalscape chart

The following diagram illustrates the anatomy of Goalscape's radial chart, showing how weighted segments, progress shading, and hierarchical nesting combine into a single cohesive visualization:

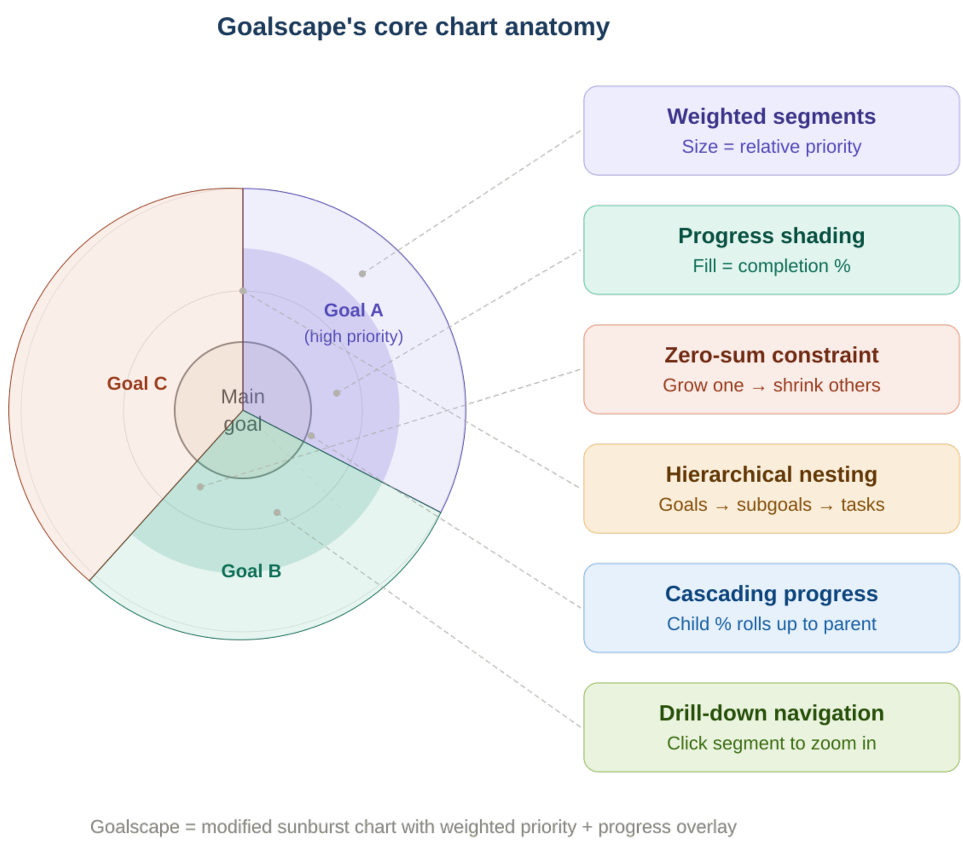

Figure 1: Goalscape's core chart anatomy — a modified sunburst with weighted priority and progress overlay

The six-core mechanics identified in Figure 1 are detailed in the table below:

Mechanic	Description	Significance
Weighted angular segments	Each slice's angular width represents its relative importance (priority), not a fixed data value. Users drag segment borders to resize.	Encodes subjective human judgment directly into the visual structure.
Zero-sum constraint	When one segment grows, its siblings shrink proportionally. The total always equals 360°.	Models the real-world constraint that time, attention, and resources are finite. Forces explicit trade-off decisions.
Progress shading	Each segment fills inward with darker shading as progress increases from 0–100%. Users adjust via a slider or by marking tasks complete.	Progress becomes a visual property of the chart, not a number in a cell. Bottlenecks are visible at a glance.
Cascading aggregation	A parent goal's progress is the weighted average of its children's completion. This cascades from leaf tasks to the center.	The center of the chart is always a truthful summary of overall project health.
Drill-down navigation	Clicking a segment zooms the view so that segment becomes the new full circle, revealing its children as the new ring.	Enables seamless transition between strategic overview and tactical detail within one interface.
Color coding	10 colors with 10 themes. Colors are categorical, not data-mapped.	Quick visual grouping without semantic overload.

2.3 What makes it work

The Goalscape chart succeeds because it exploits three cognitive advantages simultaneously:

- Preattentive processing: angular size, color, and fill darkness are all processed by the visual system before conscious attention. Users perceive priorities and problems before they read any text.
- Gestalt completeness: the circle as a bounded whole creates a natural sense of "this is everything." Unlike a scrollable list, nothing is hidden below the fold.
- Proportional honesty: the zero-sum constraint prevents the common planning fallacy of treating everything as equally important. The chart forces prioritization as a structural act, not a label.

3. The Constellation Map Concept

3.1 Definition

The **Constellation Map** is a proposed interactive radial visualization for Constellation that displays the structure, density, and maturity of a user's knowledge base as a multi-ring chart. It adapts Goalscape's chart mechanics from goal management to knowledge management, redefining the semantic meaning of each visual dimension.

3.2 Semantic mapping: from goals to knowledge

The Constellation Map reinterprets each of Goalscape's visual dimensions for knowledge management. The following diagram illustrates this transformation — the same radial paradigm, applied to a fundamentally different domain:

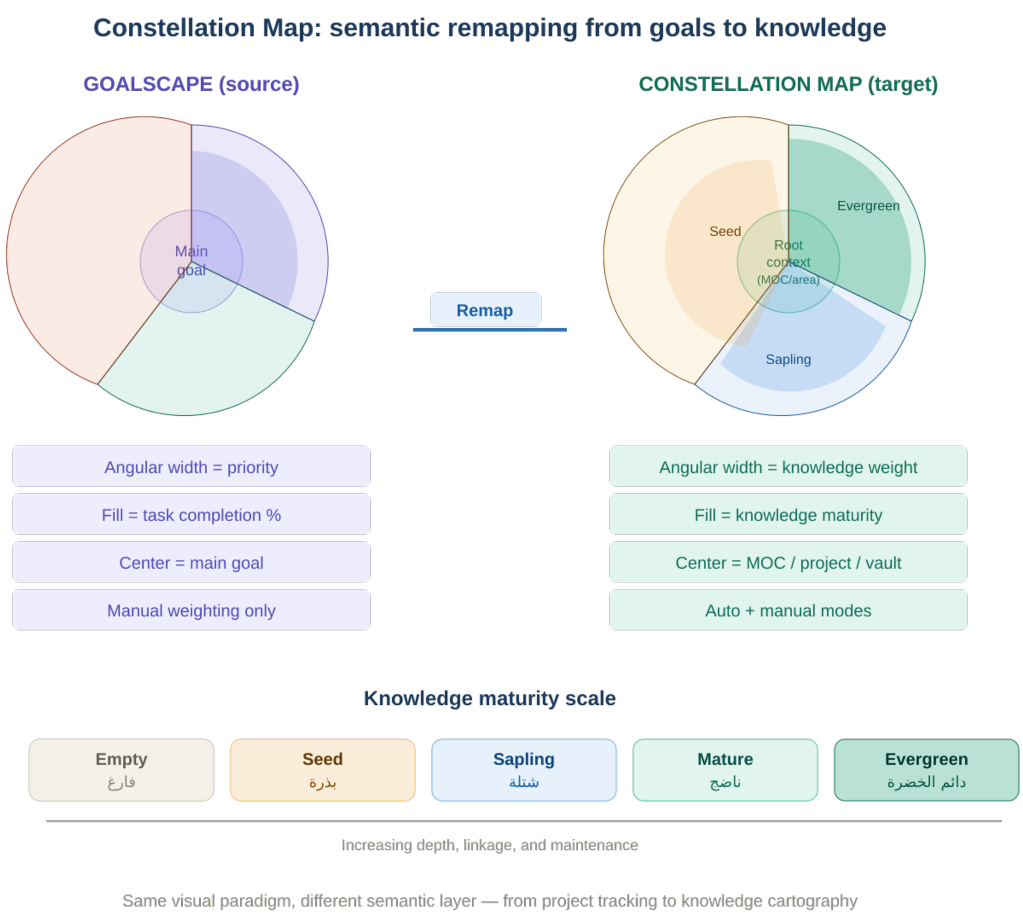

Figure 2: Semantic remapping — from goal management to knowledge cartography, with bilingual maturity scale

The detailed mapping is specified in the table below:

Goalscape dimension	Goalscape meaning	Constellation Map meaning
Center circle	Main goal	Root context: a project, MOC, area, or entire universe
Concentric rings	Hierarchy depth (goal → subgoal → task)	Knowledge depth (area → topic → note → sub-note)
Angular width	Relative importance (user-assigned)	Knowledge weight (auto-computed or user-assigned)
Fill shading	Progress toward completion (0–100%)	Knowledge maturity (seed → sapling → evergreen)
Color	Categorical grouping	Categorical grouping (tags, areas, or custom)
Drill-down	Zoom into subgoal as new center	Zoom into topic/MOC as new center
Zero-sum constraint	Priority trade-offs	Optional: for project planning mode only

3.3 Knowledge weight: the angular dimension

Angular width in the Constellation Map represents "knowledge weight" — how substantial a branch of knowledge is relative to its siblings. This can be computed in two modes:

Auto mode (default): The Rust-side Universe/Library Index computes weight automatically from structural signals. The formula combines note count (how many notes exist in the subtree), link density (how interconnected the notes are), word count (total content volume), and recency (how recently notes were created or modified). The exact weighting of these factors is configurable. Auto mode requires zero user effort and provides an honest mirror of where a user's knowledge is actually concentrated.

Manual mode (project planning): For project-oriented use cases, the user can switch to manual weighting — dragging segment borders exactly as in Goalscape. This enables the zero-sum constraint and transforms the Constellation Map into a project allocation tool. Manual mode is ideal for research planning, book outlining, or any context where the user needs to decide how much effort to devote to each branch.

3.4 Knowledge maturity: the fill dimension

Rather than a numeric 0–100% slider, the Constellation Map uses a qualitative maturity model aligned with the Digital Garden paradigm:

Level	Label (EN)	Label (AR)	Fill %	Visual
0	Empty	فارغ	0%	No fill — placeholder or stub note
1	Seed	بذرة	25%	Light fill — rough idea, a few sentences
2	Sapling	شتلة	50%	Medium fill — developing note with structure
3	Mature	ناضج	75%	Strong fill — well-developed, linked content
4	Evergreen	دائم الخضرة	100%	Full fill — authoritative, regularly maintained

Maturity can be set manually via frontmatter (e.g., maturity: sapling) or inferred automatically from note characteristics (word count thresholds, link count, last-modified recency, presence of structured headings).

The cascading mechanic from Goalscape applies: a parent topic's maturity is the weighted average of its children's maturity levels. This means the center of the chart always reflects the overall development state of the knowledge domain.

4. Design Principles

The Constellation Map is governed by five principles that distinguish it from both Goalscape's goal-centric model and generic data visualization:

Principle 1: Knowledge is the center, not tasks

The chart represents understanding, not to-dos. A fully filled segment means "I know this deeply," not "I completed this task." This aligns with PKM philosophy where knowledge is cultivated, not consumed.

Principle 2: Structure emerges from content

In auto mode, the chart's shape is a consequence of the user's actual notes, not a structure imposed before writing. The user writes; the map reveals the shape of what they've written. This respects the bottom-up, emergent ethos of Zettelkasten and Digital Gardening.

Principle 3: Gaps are features, not bugs

An empty or thin segment is not a failure — it is information. It tells the user: “You haven’t explored this area yet.” The Constellation Map makes gaps visible as a deliberate design choice, encouraging exploration rather than guilt.

Principle 4: Direction-neutral by default

The radial chart is inherently direction-neutral — it has no left-to-right or right-to-left axis. This makes it one of the few visualizations that works identically for Arabic and English content, aligning with Constellation’s first-class RTL commitment. Labels within segments follow the user’s language direction; the chart structure itself is culturally agnostic.

Principle 5: Two modes, one chart

Auto mode (mirror) and manual mode (planner) share the same visual language but serve different cognitive needs. The user can switch between “show me what I have” and “show me what I intend” within the same interface, without learning a new tool.

5. Use Cases

5.1 Knowledge landscape audit

A researcher opens the Constellation Map centered on their entire Universe. The chart reveals that 60% of their knowledge weight is concentrated in two areas (e.g., AI/ML and Systems Engineering), while several areas they consider important (e.g., Economics, Philosophy) are thin slivers with low maturity. This prompts deliberate knowledge diversification.

5.2 Book or thesis planning

An author centers the map on a book project MOC and switches to manual mode. They divide the circle into chapters, drag borders to allocate proportional effort, and track maturity as each chapter develops from seed to evergreen. The cascading fill gives an instant progress dashboard without a separate project management tool.

5.3 Research gap identification

A student studying Islamic intellectual heritage centers the map on a “Kalam and Falsafa” MOC. The chart shows strong coverage in Al-Ghazali and Ibn Sina but nearly empty segments for Ibn Rushd and contemporary analytical theology. The gap is immediately visible without reviewing individual notes.

5.4 Project decomposition and tracking

A developer centers the map on a project (e.g., "Constellation v1.0 Release"). Subgoals (GraphMind, Bases, Markdown Engine, IPC Layer) are represented as segments with manually adjusted weights reflecting their relative priority. As tasks are completed and notes updated, the maturity fill cascades upward, showing which components are on track and which lag.

5.5 Arabic/bilingual knowledge mapping

A user maintaining a bilingual universe with both Arabic and English notes sees their knowledge distributed radially. Arabic-language notes on Arab folk astronomy, Islamic jurisprudence, and Arabic linguistics appear alongside English-language technical notes. The radial layout treats both language streams equally, without the directional bias inherent in linear views.

6. Integration Architecture

6.1 Position within Constellation

The Constellation Map is a view mode, not a separate module. It sits alongside the existing views in Constellation's architecture:

View	Engine	Purpose
Editor	CodeMirror 6 + Lezer	Read and write individual notes
Graph (GraphMind)	Pixi.js (WebGL)	Force-directed relationship visualization
Bases	Custom Svelte 5	Structured database views of universe metadata
Constellation Map	SVG via D3.js or Canvas via Pixi.js	Radial knowledge density and maturity visualization

The user accesses the Constellation Map via a toggle in the view switcher, or by right-clicking a note/MOC and selecting "Open as Constellation Map."

6.2 Data pipeline

The Constellation Map consumes data from the existing Rust-side Universe Index via the IPC layer. No new data infrastructure is required. The pipeline is:

1. Universe Index (Rust): maintains the note graph, wikilink registry, and metadata index. Computes subtree statistics (note count, link density, word count, recency) on demand.

2. IPC bridge (Tauri v2 commands): exposes a constellation_map_data command that accepts a root note ID and depth limit, returning a hierarchical JSON structure with weight and maturity values per node.
3. Svelte 5 component: receives the JSON tree and renders it as an interactive radial chart. Handles drill-down navigation, mode switching, and label rendering (including RTL text shaping).

6.3 Rendering technology recommendation

SVG via D3.js is recommended over Canvas/Pixi.js for this specific view. The reasoning:

- The Constellation Map deals with dozens to low hundreds of segments, not the thousands of nodes that justify GraphMind's WebGL approach.
- SVG provides native text rendering (critical for Arabic text shaping and bidirectional labels), built-in event handling per element, and CSS-based theming.
- D3's d3-hierarchy module with d3.partition() provides sunburst layout computation, handling angular positioning and size calculation automatically.
- SVG integrates naturally with Constellation's existing CSS theming system for light/dark mode.

For universes with extremely large hierarchies (1000+ segments visible simultaneously), a **Canvas fallback** using Pixi.js can be implemented later as an optimization. The data pipeline remains identical; only the rendering layer changes.

7. Competitive Differentiation

No major PKM application currently offers a radial knowledge visualization. The following table maps the competitive landscape as of April 2026:

Application	Graph type	Supports weight/priority	Supports maturity	Radial view
Obsidian	Force-directed (2D/3D)	No	No	No
Logseq	Force-directed	No	No	No
Notion	None native	No	No	No
Roam Research	Force-directed	No	No	No
Capacities	None	No	No	No

Tana	Supertags graph	No	No	No
Goalscape	Radial (sunburst variant)	Yes	Yes (as progress)	Yes
Constellation (proposed)	Radial + force-directed	Yes (auto + manual)	Yes (Digital Garden model)	Yes

The Constellation Map would make Constellation the first PKM application to combine a force-directed graph (for relationship exploration) with a radial chart (for structural overview). These two views serve complementary cognitive purposes: the graph answers “what is connected to what?” while the radial map answers “where is my knowledge concentrated and how mature is it?”

Naming synergy

The name “Constellation” itself evokes a radial, star-map metaphor. The five-node Universe Mark logo echoes a central node with orbiting connections. A radial knowledge visualization is not merely a feature addition — it is the fulfillment of the product’s visual identity.

8. Risks and Mitigations

Risk	Severity	Mitigation
Visual clutter at deep hierarchy levels	Medium	Limit visible depth to 3–4 rings by default. Deeper levels accessible only via drill-down. Collapse small segments into an “Other” arc.
Performance with large universes (10,000+ notes)	Medium	Compute weights lazily in Rust, cache results. Limit initial render to top 2–3 levels. Use virtual rendering for outer rings.
User confusion between auto and manual modes	Low	Clear visual indicator (e.g., lock icon on segments in manual mode). Default to auto mode; manual mode requires explicit activation.
Arabic label rendering in arc segments	Low	Use straight labels with leader lines rather than curved text along arcs. Arabic text shaping is handled by the existing RTL infrastructure.
Conceptual overlap with GraphMind	Low	Position them as complementary views: GraphMind = relationship topology, Constellation Map = structural overview. Different questions, different answers.

9. Suggested Development Roadmap

The Constellation Map can be developed in three incremental phases, each delivering standalone value:

Phase 1: Static universe overview (foundation)

- Implement the Rust-side IPC command for subtree weight computation.
- Build the Svelte 5 radial chart component using D3.js d3.partition().
- Render a read-only universe-level Constellation Map with auto-computed weights.
- Support basic interactions: hover for note metadata, click to navigate to note.
- Deliverable: a “knowledge bird’s-eye view” users can open from any MOC.

Phase 2: Maturity and drill-down (enrichment)

- Implement maturity inference from note characteristics (or read from frontmatter).
- Add fill shading that reflects maturity levels per segment.
- Implement drill-down navigation (click segment to re-center).
- Add breadcrumb trail showing drill-down path.
- Deliverable: an interactive knowledge exploration tool.

Phase 3: Manual mode and project planning (expansion)

- Implement draggable segment borders for manual weight assignment.
- Enable zero-sum constraint toggle.
- Store manual weights in note frontmatter or a project configuration file.
- Integrate with Bases: allow project-type database views to generate a Constellation Map.
- Deliverable: a dual-mode tool that serves both knowledge mapping and project planning.

10. Conclusion

Goalscape demonstrates that a radial chart with weighted segments, progress shading, and hierarchical drill-down can transform how people relate to complex, multi-level information. Its success in goal management — adopted by Olympic athletes, multinational corporations, and government agencies — validates the cognitive power of this visualization paradigm.

The Constellation Map proposes to transplant this paradigm into personal knowledge management, redefining its dimensions from goals and progress to knowledge and maturity.

The result would be a feature unique in the PKM landscape: a visual answer to the question "What does the shape of my knowledge look like?"

This concept aligns with Constellation's identity at every level — from its name and logo to its technical architecture (Rust-computed graph data, Svelte 5 rendering, first-class RTL support). It complements rather than competes with GraphMind, offering a structural overview where the graph offers relational exploration. And it can be built incrementally, delivering value at each phase.

Next steps

1. Review and approve this concept paper. 2. Create a detailed technical specification for Phase 1, including IPC contract, D3.js component architecture, and Svelte 5 integration points. 3. Build a standalone prototype of the radial chart component to validate rendering performance and interaction design. 4. Integrate the prototype into Constellation's view switcher.

End of concept paper — Constellation Map v1.0